

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2024-2025

Comprehensive Exam
(EC-3 Regular)

Course No. : AIMLCZG512
Course Title : Deep Reinforcement Learning
Nature of Exam : Closed Book
Weightage : 40%
Duration : 2 Hours
Date of Exam : 30-March-2025 (EN)

No. of Pages	= 4
No. of Questions	= 6

Note to Students:

1. Please follow all the *instructions for candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1)

[2+5 = 7 Marks]

- (a) Consider a reinforcement learning scenario: A self-driving car learns to navigate a city using reinforcement learning. The vehicle receives a reward of +100 for reaching its destination, -10 for stopping at a red light, and -1 for every second it spends driving. The agent's objective is to maximize the discounted cumulative reward.

Given this scenario, explain the role of the discount factor briefly [1.0 M]. If the γ is increased from 0.9 to 0.99, how would this change the self-driving car's decision-making in this scenario? Justify your answer. [1.0 M].

- (b) A gambler plays a 4-armed bandit (slot machine with 4 levers) using the ϵ -greedy strategy with $\epsilon=0.2$. The estimated values (Q-values) of the four arms at a given time step are:

$$Q_1 = 3.5, \quad Q_2 = 5.0, \quad Q_3 = 4.2, \quad Q_4 = 6.1$$

(b-1) What is the probability of selecting the optimal arm? [1.0 M].

(b-2) What is the probability of choosing each of the other arms? [1.0 M].

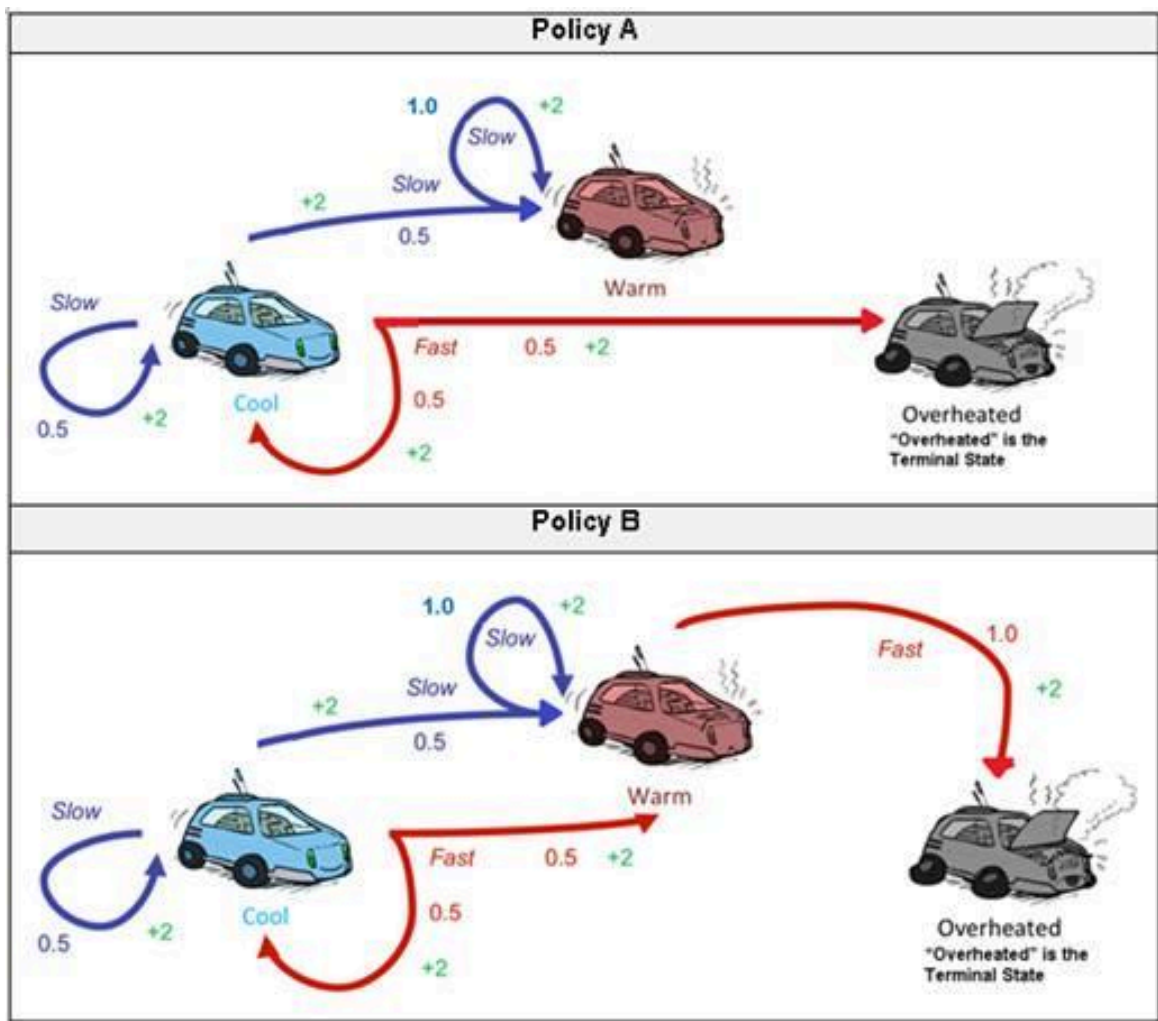
(b-3) Suppose the gambler chooses arm 2 and receives a reward of 7.0. Using the incremental update rule with a step size of $\alpha=0.1$, calculate the updated Q_2 value. [1.0 M].

(b-4) Compute the **expected reward at this step**, assuming the agent follows the ϵ -greedy policy and the true expected rewards of the arms are: $R_1 = 4.0$, $R_2 = 5.5$, $R_3 = 4.5$, $R_4 = 6.0$. [2.0 M].

Q2)

[3+2+2= 7 Marks]

- (a) A racecar is learning to maximize the discounted cumulative reward while learning to drive effectively on a track. Two different policies (Policy A and Policy B) designed are shown below in state transition diagrams. Use $\gamma = 1$ and constant transition reward = +2.



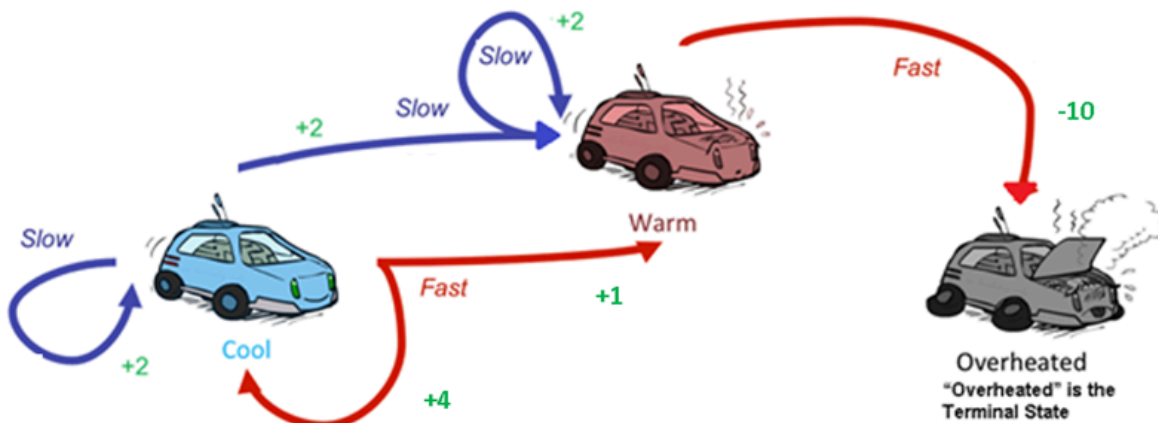
Compare & comment on the suitability of the given policies A and B w.r.t dynamic programming. Justify your comment for each using no more than four well-defined statements. [3.0 M]

- (b) Using policy A from above, apply asynchronous value iteration in this specific order = (V(Warm), V(Overheated), V(Cool)) for three iterations. Assume $V(S_0) = 6$ for all the states, including the terminal state. [2.0 M]
- (c) Observe the results of the three iterations and analyze if a thresholding of $\Delta \leq 2$ is applied for all the states in the algorithm, will the algorithm converge for the given set-up? Justify your answer in no more than two well-defined statements. **Note** : After iteration t , for every state "S", $\Delta_t(S) = |V_{t+1}(S) - V_t(S)|$ is the difference between the value of the state between subsequent iterations [2.0 M]

Q3)

[4+1+2= 7 Marks]

Consider the policy of an RL agent given below to answer the questions



- (a) Apply on policy first visit Monte Carlo control find the value of the state “Cool” using below episode with $\gamma = 0.5$ and $\epsilon = 0.6$. [4.0 M]

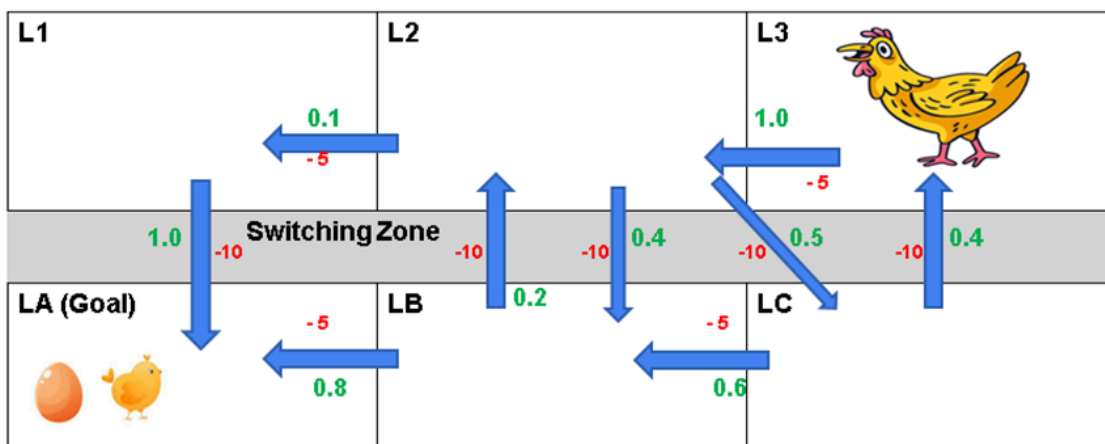
Episode : (Cool, Slow, Cool, Slow, Cool, Fast, Warm, Fast, Overheated)

- (b) If the above problem is applied with an off-policy MC algorithm, justify the significance of weighted importance sampling compared to ordinary importance sampling. [1.0 M]
- (c) Explain a scenario where Temporal Difference learning becomes equivalent to Monte Carlo learning if applied to the given use case. [2.0 M]

Q4)

[5+2 = 7 Marks]

Consider the following grid-based environment where a chicken navigates through different locations (L1, L2, L3, LA, LB, and LC) to reach the goal state (LA). Movement is only permitted between states along the blue arrows. Each arrow (transition between states) is marked with its corresponding rewards (in red colour) and the probability of taking that action (in green colour). You can observe that the transitions leading to crossing the "Switching Zone" result in higher penalties. You can observe that the policy is implied by the probability of action (number written in green) for each state.



- (a) Apply Q-Learning for one iteration with $\gamma = 1$ and $\alpha = 0.2$. In an episode, use greedy action selection per time step. [5.0 M]

- (b) Assume that the chicken uses the policy where the action with the greatest Q-value is always preferred. Is this an optimal policy? Why or Why not? Explain in no more than three well-formed sentences. [2.0 M]

Q5)

[4 + 3 = 7 Marks]

- a) Consider the following scenarios:

Scenario 01: Tic-Tac-Toe game environment with a small, finite state space.

Scenario 02: Autonomous driving in a real-world city with a large, continuous state space.

Answer the following questions:

- a-1. Which methods (Tabular or Function Approximation) are appropriate for the given scenarios? Justify your selection. [1.0 M]
- a-2. Why are Semi-Gradient methods preferred over Stochastic Gradient methods for value function learning in continuous spaces? Provide justification. [1.0 M]
- a-3. Linear function approximators often require handcrafted, domain-dependent features, which depend on expert knowledge. Why is this considered a limitation? How do deep learning-based function approximators address this issue through automatic feature extraction? Mention two techniques used in deep learning for automatic feature extraction. [2.0 M]
- b) Answer the following
- b-1. REINFORCE and REINFORCE with baseline are both policy gradient methods. What is the key difference between REINFORCE and REINFORCE with baseline, and how does the baseline help in reducing variance? [1.0 M]
- b-2. Imagine a robot navigating a maze to reach a goal. [2.0 M]
- a. What is the role of the actor and critic r in the Actor-Critic method for the robot navigation scenario?
- b. Why is it typically impractical to run policy evaluation and policy improvement steps to convergence in large-scale reinforcement learning problems?

Q6)

[3 + 2 = 5 Marks]

- (a) How AlphaGoZero with a single network outperforms AlphaGo with dedicated network for policy & value? Explain the role of MCTS in both the RL programs. [3.0 M]
- (b) Explain why DAGGER is a reinforcement learning algorithm with two well-articulated points [2.0 M]